María Teresa Muñoz Martín

Machine Learning Engineer · LLM & GenAI · NLP Engineer

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Computer Engineer specialising in **Large Language Models**, **Generative AI** and **Natural Language Processing**, with end-to-end experience across the full LLM lifecycle: **fine-tuning** (LoRA, QLoRA), **model quantisation** (NF4, FP4, Int8, bitsandbytes) for efficient and sustainable inference (**Green AI**), production-grade **Retrieval-Augmented Generation** (hybrid search, cross-encoder re-ranking, RAGAS evaluation), **prompt engineering** and **LLM evaluation** (RAGAS, lm-evaluation-harness, LLM-as-a-judge). Proven research-to-production track record on low-resource language technology, validated PEFT and quantisation protocols enabling Basque LLMs (*Latxa-3.1-8B*) to run on consumer-grade GPUs, and clinical-domain GenAI applications (RAG chatbot for eating-disorder prevention, awarded 9.8/10). Currently contributing to **ALIA**, Spain's sovereign-AI initiative for Spanish LLMs, and to two European projects on multilingual hate-speech detection and counter-narrative generation. Co-author of an accepted **SemEval-2026 / ACL 2026** paper on knowledge distillation. Committed to responsible, human-in-the-loop AI.

Work Experience

Machine Learning Engineer — ALIA Project
Jan 2026 – Present
Jaén, Spain

SINAI Research Group, University of Jaén - UJA

Spanish Government strategic initiative for sovereign Spanish LLMs.

- Engineered **multi-million-token** Spanish corpus-curation pipelines for retrieval-augmented LLM training, integrating deduplication, language identification and toxicity filtering at scale.
- Delivered high-precision Spanish↔English translation and quality-control workflows producing benchmark-ready parallel corpora for cross-lingual LLM evaluation.
- Defined model-evaluation protocols aligned with **EU AI Act** transparency requirements and responsible-AI standards (bias and toxicity probes).
- Contributing to **RomaNET** (European Commission, *CERV-2024-CHAR-LITI-SPEECH*): multilingual NLP pipelines for hate-speech and hate-crime detection targeting anti-Roma discourse.
- Building a web platform for **LLM-driven counter-narrative generation** against antigypsyism with the National Federation of Roma Women's Associations (*KAMIRA*); prompt engineering, retrieval grounding and content-safety guardrails.
- Curated and annotated specialised Spanish corpora for **argument mining** and **stance detection**; designed annotation guidelines, double-annotation protocols and inter-annotator agreement quality controls.
- Contributed to three competitive research projects (*CONSENSO*, *MODERATES*, *SocialTOX*) producing publication-ready datasets and trained models.

Generative AI & LLM Engineer — Investigo Programme (NextGenerationEU)
Nov 2024 – Oct 2025
San Sebastian, Spain

HITZ Center for Language Technology, University of Basque Country - EHU

- Designed and benchmarked a Parameter-Efficient Fine-Tuning protocol for Basque LLMs, conducting a rigorous comparative analysis of the integrated **QLoRA** workflow against the standard **LoRA-Q** approach (full-precision adaptation followed by post-hoc compression) across **Int8, FP4 and NF4** precisions.
- Evaluated three architectures — generalist **Gemma-2-9B**, **Qwen-3-8B** and the Basque-specialised **Latxa-3.1-8B**; established **NF4 with Double Quantization** as the most robust configuration, consistently outperforming integer-based formats in linguistic quality.
- Achieved >**50% VRAM reduction**, enabling 8B-parameter LLMs to run on **consumer-grade GPUs (RTX 3090/4090)** instead of datacentre infrastructure; quantified the spatial-efficiency vs. inference-latency

Technical Skills

Programming
Python · SQL / MySQL · C / C++ · Java

LLM & GenAI
LoRA · QLoRA · NF4 · FP4 · Int8 · Double Quantization · bitsandbytes · vLLM · Knowledge Distillation · Prompt Engineering · lm-evaluation-harness · RAGAS · LLM-as-a-judge · Responsible / Green AI

RAG & Retrieval
Hybrid Search (BM25+dense) · Cross-encoder Re-ranking · Reciprocal Rank Fusion · Query Rewriting / HyDE · Semantic Chunking · RAGAS Evaluation

NLP Tasks
Argument Mining · Stance Detection · Hate-Speech Detection · Multilingual / Low-resource NLP · Cross-lingual Evaluation

Frameworks
PyTorch · Transformers · LangChain · LlamaIndex · FastAPI · Ollama · React / Next.js · W&B

Infrastructure & MLOps
ChromaDB · Qdrant · HuggingFace Hub · GitHub · GitHub Actions · Vercel · Docker / docker-compose · HPC / GPU Clusters · LaTeX

trade-off and engineered distributed inference pipelines on HPC/GPU clusters with **vLLM**, defining a validated protocol for **sustainable Green AI** in low-resource languages.

Research Assistant — Initiation to Research Grant
Sep 2023 – May 2024
Jaén, Spain

SINAI Research Group, University of Jaén - UJA

- Studied and implemented early **Retrieval-Augmented Generation (RAG)** pipelines, including dense passage retrieval with **FAISS**, sentence-level **embeddings** (Sentence-BERT), and prompt-constrained generation over **Transformer**-based LLMs (GPT-3.5, LLaMA-2); explored chunking strategies, vector-store indexing, and context-window management for domain-specific Q&A systems.

NLP Research Intern — Ícaro Programme
Oct 2023 – Mar 2024
Jaén, Spain

SINAI Research Group, University of Jaén - UJA

Applied research in NLP/ML; supported retrieval-augmented systems and corpus-engineering pipelines.

Education

Erasmus Mundus M.Sc. — Language & Communication Technologies
2024 – 2026

University of Basque Country - EHU (Spain) & University of Groningen - RUG (Netherlands)

Erasmus Mundus Scholar

Thesis: *QLoRA vs LoRA-Q Pipelines for Basque: a Multi-Model Efficiency Analysis.*

B.Sc. in Computer Engineering
2020 – 2024

University of Jaén - UJA

GPA: **8.22 / 10** | **Erasmus+ exchange** at KTU, Lithuania (2023–24)

Thesis: *BoTCA: RAG chatbot for eating-disorder prevention.* **9.8 / 10**

Technological Baccalaureate
2018 – 2020

IES Santa Catalina de Alejandría, Jaén

Erasmus+, Rome 2019 **1 of 2 students** selected from 150 candidates.

Publications & Research

Sylloscope at SemEval-2026 Task 11: *Decoupling Logic from Belief via DeepSeek-Enhanced Distillation in Qwen Models*

*SemEval-2026 Workshop Task 11, co-located with **ACL 2026***

Designed a novel pipeline combining **DeepSeek-enhanced knowledge distillation** with Qwen models to decouple logical reasoning from belief-based inference in stereotyped-claim detection. Achieved **96.86% accuracy** (ranking score 39.81) on Subtask 1 and **91.67% accuracy** (ranking score 26.02) on Subtask 3. Code: github.com/SemHuis/group6-shared-task

Performant Lightweight Encoders for Spanish in the Legal and Administrative Domains

Journal of Natural Language Processing (SEPLN 2026), Issue 77

Introduced two lightweight, domain-adapted models (bi-encoder and cross-encoder) for Spanish legal and administrative information retrieval built on the MrBERT-es backbone. Utilized a synthetic data generation pipeline combined with a six-phase curriculum learning strategy and positive-aware mining. Outperformed larger general-purpose models like bge-m3 and Qwen3-Embedding on specialized benchmarks, demonstrating that targeted domain adaptation and careful training design can effectively compensate for model scale.

Certifications

LangChain — *Introduction to LangGraph - python*

Microsoft — *Applied Skills: Create a Generative AI Chat App*

Hugging Face — *Agents Course*

Anthropic — *AI Fluency: Framework & Foundations*

Anthropic — *Claude Code in Action*

Anthropic — *Introduction to Model Context Protocol (MCP)*

Languages

Spanish	Native
English	Full Professional (Cambridge B2)
Driving licence	Category B

Soft Skills

- **Cross-cultural adaptability:** research & study experience in Spain, Italy, Lithuania, Netherlands.
- **Research mindset:** design experiments, build benchmarks from scratch, iterate fast.
- **Ethical commitment:** human-in-the-loop and responsible-AI advocate.
- **Discipline:** competitive athlete (skiing, swimming, weightlifting).

References

Eugenio Martínez Cámara — emcamara@ujaen.es
Bachelor's Thesis Supervisor, UJA

Aitor Soroa Echave — a.soroa@ehu.eus
Master's Thesis Supervisor, EHU

Germán Rigau Claramunt — german.rigau@ehu.eus
Master's Thesis Supervisor, EHU

Rik van Noord — r.i.k.van.noord@rug.nl
Master's Thesis Supervisor, RUG

Personal Projects

Gender Gap in Open-Source AI — gender-gap-oss.vercel.app

Full-stack data platform measuring gender representation in open-source AI: surfaced **5%** women in the top 500 GitHub accounts and **7.4%** across contributors to 20 major AI repositories (PyTorch, Transformers, LangChain, JAX, etc.). Built a tiered, fully **offline gender-inference pipeline** (GitHub GraphQL pronouns, bio parsing, name dictionaries) with no third-party gender-API calls. Automated refresh via **GitHub Actions** (monthly + daily) with continuous deployment on **Vercel**. *Stack: Python, GitHub REST/GraphQL, Next.js, Tailwind, D3.js.*

PergamIA — Multi-tenant Corporate RAG System

Production-grade **multi-tenant RAG** with **RBAC**: **hybrid search** (BM25 + dense, RRF fusion), **cross-encoder re-ranking**, query rewriting, streaming, **RAGAS** evaluation and per-query audit logs. *Stack: FastAPI, LangChain, ChromaDB, Ollama, React/Next.js, Docker.*